

STEERING

06

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(STEERING)..... 06-10-1

GENERAL PRECAUTION (STEERING)

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Wheels and Tire Installation

1. When installing the wheels and tires, tighten the wheel nuts in a criss-cross pattern to the following tightening torque.

Tightening torque

88.2—117.6 N·m {9.00—11.99 kgf·m, 65.06—86.73 ft·lbf}

Connector Disconnection

1. Disconnect the negative battery cable before performing any work that requires handling of connectors. (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)

Power Steering Components Removal/Installation

1. If any power steering fluid line has been disconnected anytime during the procedure, add ATF M-III or equivalent (e.g. Dexron®II), bleed the fluid lines, and inspect for leakage after the procedure has been completed.

06

06-14 POWER STEERING

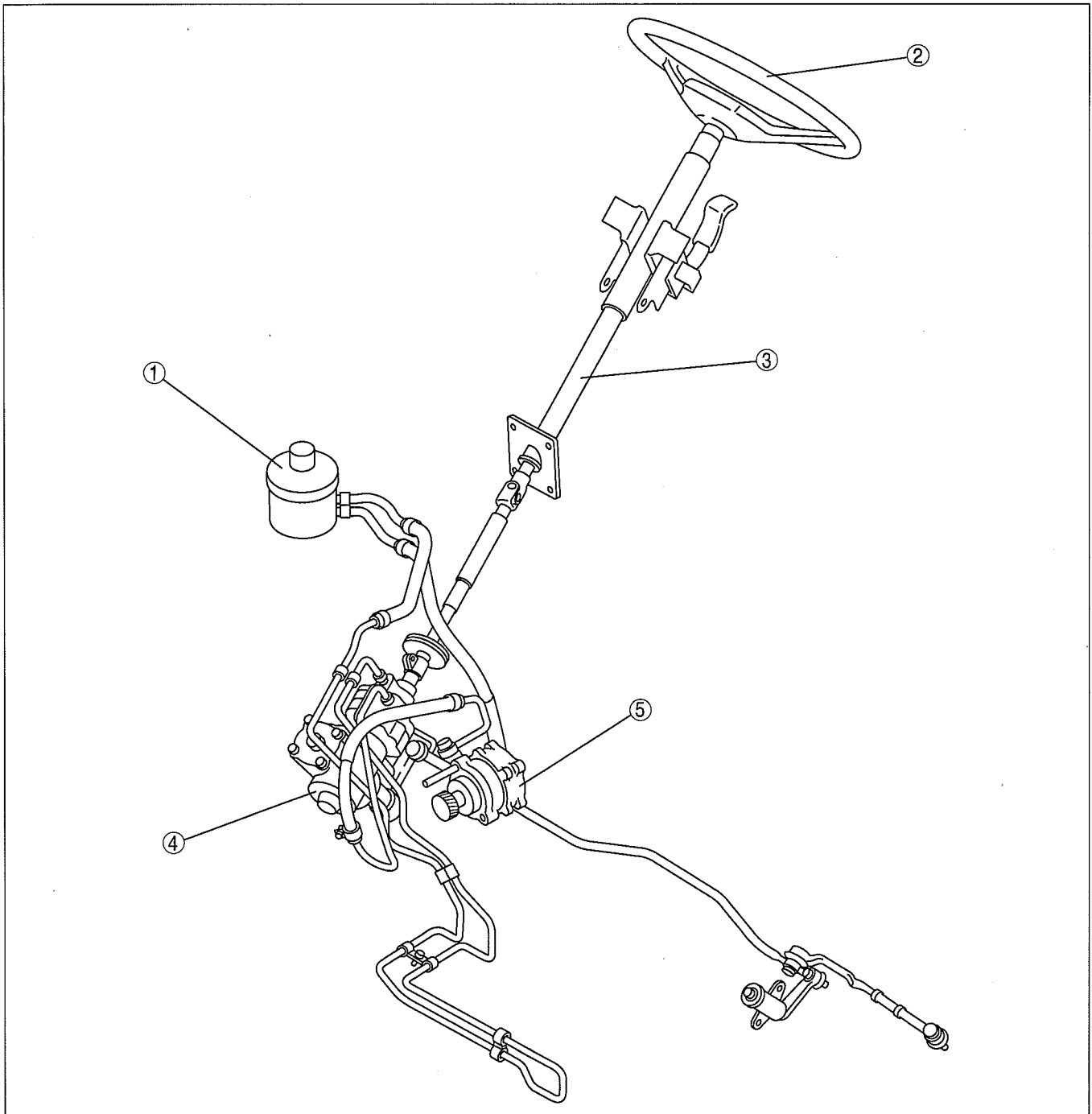
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POWER STEERING

STEERING LOCATION INDEX

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1	Power steering fluid tank (See 06-14-3 AIR BLEEDING.) (See 06-14-3 POWER STEERING FLUID INSPECTION.)
2	Steering wheel and column (See 06-14-5 STEERING WHEEL AND COLUMN INSPECTION.) (See 06-14-6 STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
3	Steering shaft (See 06-14-9 STEERING SHAFT INSPECTION.)

4	Steering gear and linkage (See 06-14-10 STEERING GEAR AND LINKAGE REMOVAL/INSTALLATION.) (See 06-14-11 STEERING GEAR AND LINKAGE DISASSEMBLY/ASSEMBLY.) (See 06-14-13 STEERING GEAR AND LINKAGE INSPECTION.)
5	Power steering oil pump (See 06-14-14 POWER STEERING OIL PUMP REMOVAL/INSTALLATION.) (See 06-14-15 POWER STEERING OIL PUMP DISASSEMBLY/ASSEMBLY.)

POWER STEERING

AIR BLEEDING

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Caution

- Do not turn the steering wheel during the fluid level inspection, otherwise the fluid level changes and cannot be inspected correctly.

1. Inspect the fluid level. (See 06-14-3 POWER STEERING FLUID INSPECTION.)
2. Jack up the front of the vehicle and support it on safety stands.
3. Turn the steering wheel fully to the left and right several times with the engine not running.
4. Reinspect the fluid level.
 - If it has dropped, add fluid.
5. Repeat Steps 3 and 4 until the fluid level stabilizes.
6. Lower the vehicle.
7. Start the engine and let it idle.
8. Turn the steering wheel fully to the left and right several times.
9. Verify that the fluid is not foamy and that the fluid level has not dropped.
 - If the fluid level has dropped, add fluid as necessary and repeat Steps 8 and 9.

POWER STEERING FLUID INSPECTION

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Fluid Level Inspection

1. Inspect the power steering fluid level.
 - Add fluid to the specified level as necessary.

Power steering fluid type

ATF M-III or equivalent (e.g. Dexron®II)

Power steering fluid capacity (approx. quantity)

1.1 L {1.2 US qt, 1.0 Imp qt}

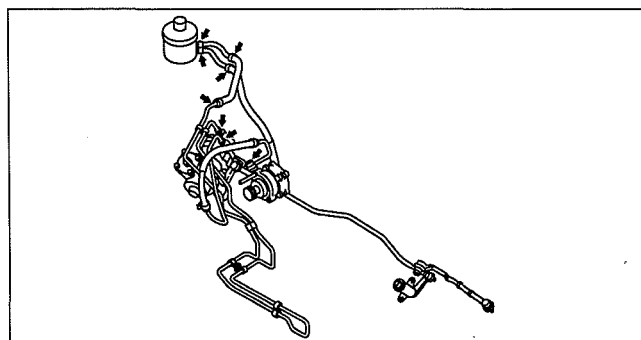
Fluid Leakage Inspection

1. Start the engine and let it idle.
2. Turn the steering wheel fully to the left and right to apply fluid pressure.

Caution

- If the steering wheel is kept in the fully turned position for more than 5 seconds, the fluid temperature will rise excessively and adversely affect the oil pump.

3. Inspect for fluid leakage at the points indicated in the figure.
 - If fluid leakage is found, replace related parts.



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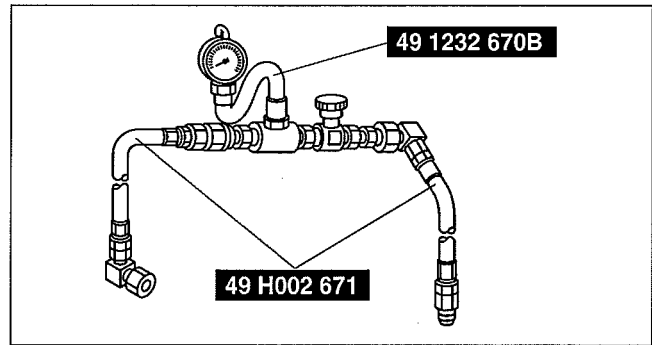
POWER STEERING

Fluid Pressure Inspection

1. Assemble the **SSTs** as shown in the figure.
2. Mark both hose connections to ensure that the hose is reinstalled in its original position.
3. Disconnect the pressure pipe from the oil pump, and connect the **SSTs**.

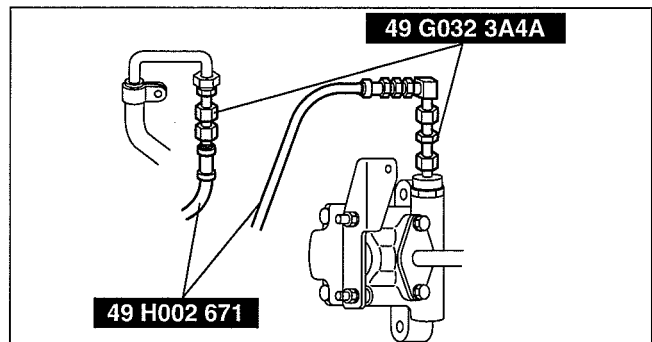
Tightening torque

29.4—44.1 N·m {3.00—4.49 kgf·m, 21.7—32.5 ft·lbf}



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4. Connect the **SSTs** (49 1232 670B and 49 H002 671) to the **SST** (49 G032 3A4A).
5. Bleed the air from the system. (See 06-14-3 AIR BLEEDING.)
6. Open the gauge valve fully.
7. Start the engine and turn the steering wheel fully left and right to raise the fluid temperature to 50—60 °C {122—140 °F}.



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Caution

- If the valve is left closed for more than 5 s, the fluid temperature will increase excessively and adversely affect the oil pump.

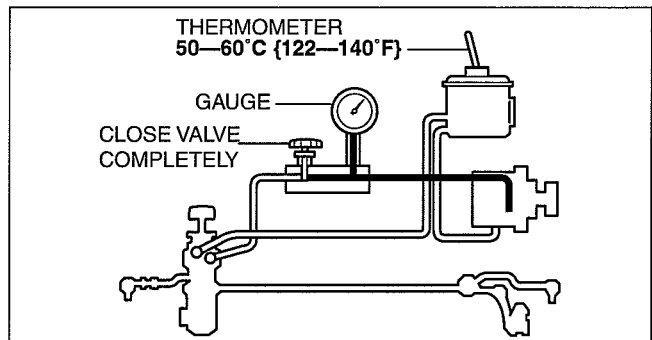
8. Close the gauge valve completely. Increase the engine speed to 1,000—1,500 rpm and measure the fluid pressure generated by the oil pump. If the pressure is not within the specification, repair or replace the oil pump component.

Oil pump fluid pressure

9.3—9.8 MPa {94.8—99.9 kgf/cm², 1348—1421 psi}

Caution

- If the steering wheel is kept in the fully turned position for more than 5 s, the fluid temperature will rise excessively and adversely affect the oil pump.



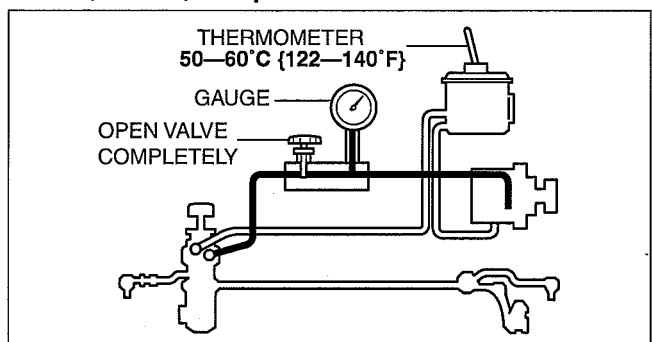
DBR614ZWB614

9. Open the gauge valve fully and increase the engine speed to 1,000—1,500 rpm
10. Turn the steering wheel fully to left and right and measure the fluid pressure generated at the gear housing. If the pressure is not within the specification, repair or replace the steering gear component.

Gear housing fluid pressure

9.30—9.79 MPa {94.8—99.8 kgf/cm², 1348—1419psi}

11. Remove the **SSTs**. Install and tighten the pressure pipe to the specified torque.



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Tightening torque

31.4—47.0 N·m {3.21—4.79 kgf·m, 23.2—34. ft·lbf}

12. Bleed the air from the system.

POWER STEERING

STEERING WHEEL AND COLUMN INSPECTION

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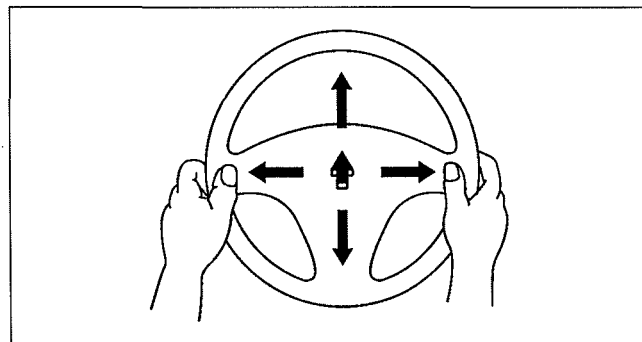
Steering Wheel Play Inspection

1. With the wheels in the straight-ahead position, gently turn the steering wheel to the left and right and verify that the play is within the specification.
 - If the play exceeds the specification, either the steering joints are worn or the backlash of the steering gear is excessive. Correct as necessary.

Steering wheel play
0—30 mm {0—1.18 in}

Steering Wheel Looseness Inspection

1. Move the steering wheel as shown in the figure to inspect for column bearing wear, steering shaft joint play, steering wheel looseness, and column looseness.
 - Repair or replace as necessary.



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Steering Wheel Effort Inspection

1. Inspect the following points:
 - Tire size and tire pressure
 - Fluid level
2. With the vehicle on a hard, level surface, put the wheels in the straight-ahead position.
3. Remove the air bag module.

Warning

- Handling the air bag module improperly can accidentally deploy the air bag module, which may seriously injure you. Read AIR BAG SYSTEM WARNINGS before handling the air bag module. (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)

4. Start the engine and warm the power steering fluid to 50—60 °C {122—140 °F}.
5. Measure the steering wheel effort using a torque wrench.

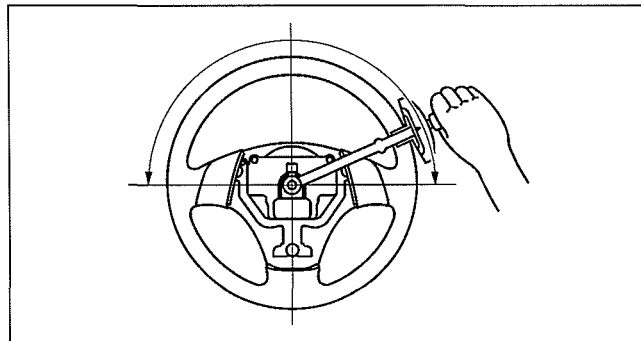
- If not within the specification, verify the following:
 - No air in steering system
 - No fluid leakage at hose or connectors
 - Function of oil pump and steering gear

Steering wheel effort

7.8 N·m {80 kgf·cm, 58 in·lbf} max.

Note

- To determine whether the steering effort is satisfactory or not, perform the inspection on another vehicle of the same model and under the same conditions, and compare the results.
- The steering wheel effort varies with conditions as shown below.
 - Road conditions, such as dry or wet, and asphalt or concrete.
 - Tire conditions, such as brand, wear, and tire pressure.



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POWER STEERING

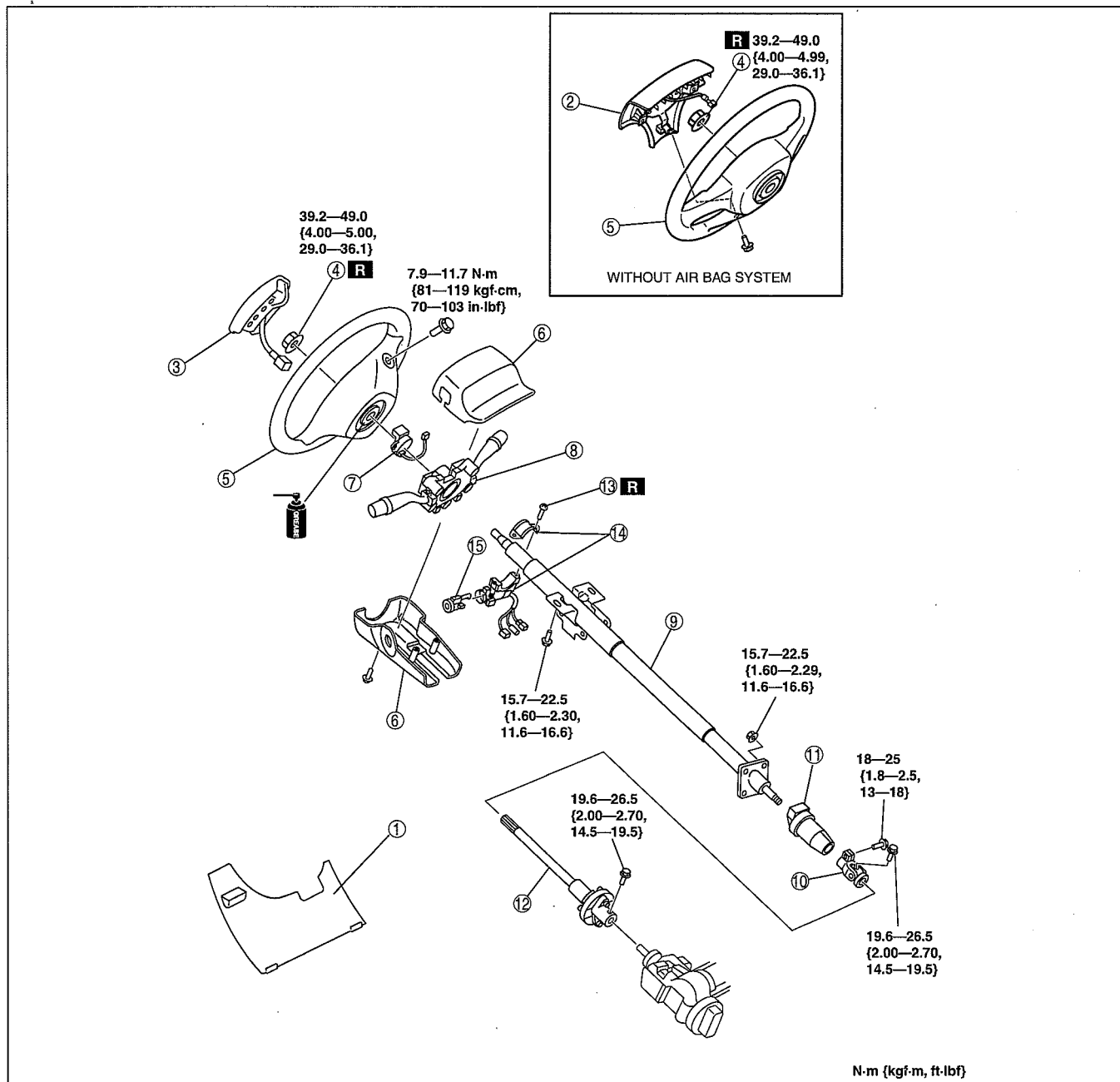
STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION

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Warning

- Handling the air bag module improperly can accidentally deploy the air bag module, which may seriously injure you. Read AIR BAG SYSTEM WARNINGS before handling the air bag module. (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)

- Remove in the order indicated in the table.
- Install in the reverse order of removal.



DBR614ZWB101

1	Lower panel (See 09-17-10 LOWER PANEL REMOVAL/INSTALLATION.)
2	Horn cap
3	Air bag module (See 08-10-4 DRIVER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION.)
4	Locknut

5	Steering wheel (See 06-14-7 Steering Wheel Removal Note.) (See 06-14-8 Steering Wheel Installation Note.)
6	Column cover (See 09-17-8 COLUMN COVER REMOVAL/INSTALLATION.)
7	Clock spring (See 08-10-7 CLOCK SPRING REMOVAL/INSTALLATION.)

POWER STEERING

8	Combination switch (See 09-18-9 COMBINATION SWITCH REMOVAL/INSTALLATION.)
9	Steering shaft (See 06-14-8 Steering Shaft Installation Note.)
10	Universal joint
11	Boot
12	Intermediate shaft

13	Steering lock mounting bolts (See 06-14-7 Steering Lock Mounting Bolts Removal Note.) (See 06-14-8 Steering Lock Mounting Bolts Installation Note.)
14	Steering lock component
15	Outer cylinder (See 06-14-7 Outer Cylinder Removal Note.)

Steering Wheel Removal Note

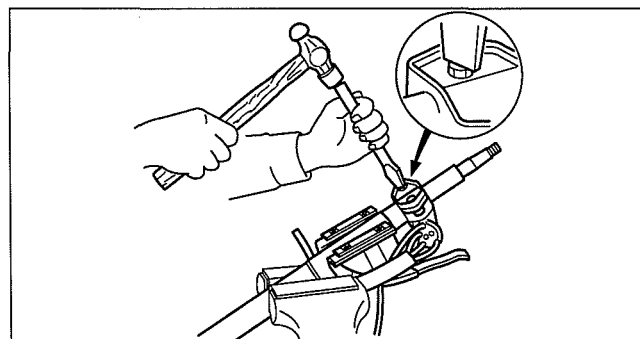
Caution

- Do not try to remove the steering wheel by hitting the shaft with a hammer. The column will collapse.

1. Set the vehicle in the straight-ahead position.
2. Remove the steering wheel using a suitable puller.

Steering Lock Mounting Bolts Removal Note

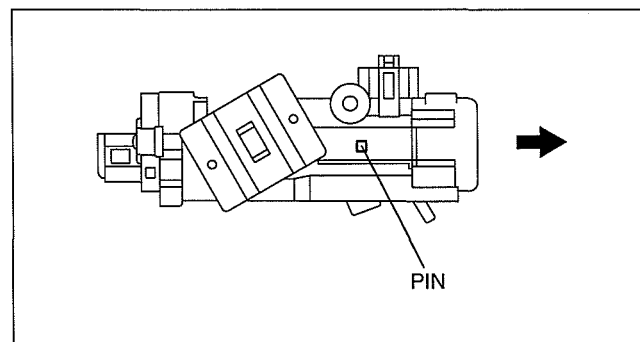
1. Make a groove in the heads of the steering lock mounting bolts using a chisel and a hammer.
2. Remove the bolts using a flathead screwdriver.
3. Disassemble the steering lock component.



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Outer Cylinder Removal Note

1. Turn the key to the ACC position.
2. Push the pin and remove the outer cylinder.



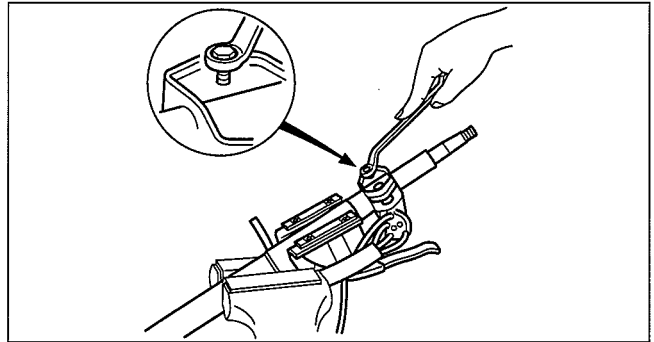
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Steering Lock Mounting Bolts Installation Note

1. Assemble the steering lock component to the steering shaft.
2. Verify that the lock operates correctly.
3. Install new steering lock mounting bolts.
4. Tighten the bolts until the heads break off.



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Steering Shaft Installation Note

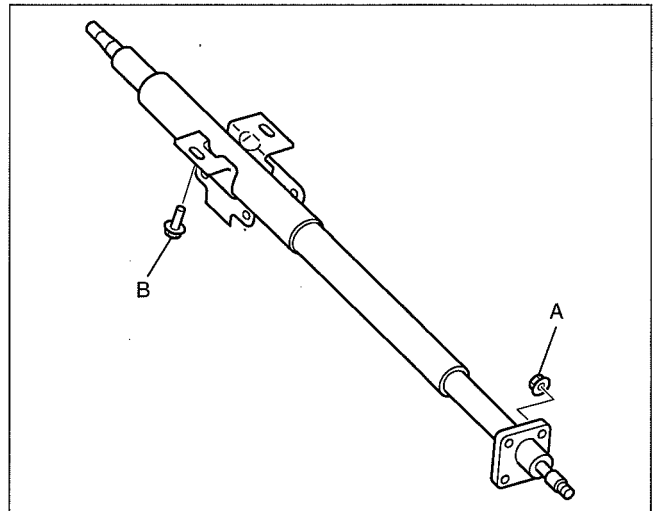
Caution

- Do not apply a shock in the axial direction of the shaft.

1. Lock the tilt lever. (Equipped with tilt mechanism)
2. Tighten nut A.
3. Tighten bolt B.

Tightening torque

15.7—22.5 N·m {1.60—2.29 kgf·m, 11.6—16.5 ft·lbf}



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Steering Wheel Installation Note

1. Make sure the wheels in the straight-ahead position, and install the steering wheel.

POWER STEERING

STEERING SHAFT INSPECTION

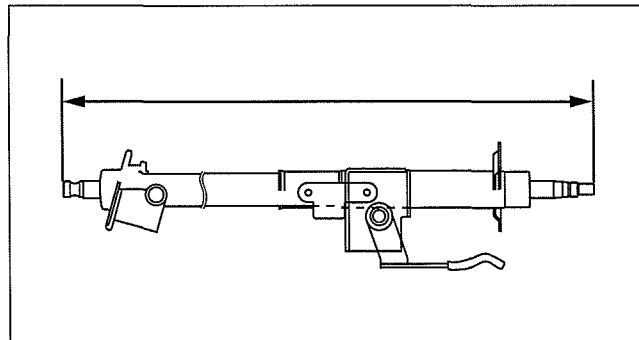
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1. Inspect the following.
 - (1) Column bearing for damage
 - (2) Steering shaft length

Steering shaft length

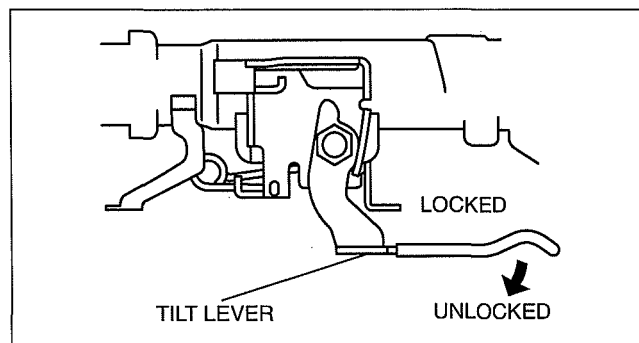
789.9—792.9 mm {31.10—31.22 in}

- (3) Dust boot damage
 - Replace the steering shaft component as necessary.



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2. Inspect the tilt operation. (Equipped with tilt mechanism)
 - (1) Verify that the tilt lever moves smoothly from unlock position to lock position.
 - (2) Verify that the steering shaft is fixed firmly when the tilt lever is locked.
 - Replace the steering shaft component as necessary.



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POWER STEERING

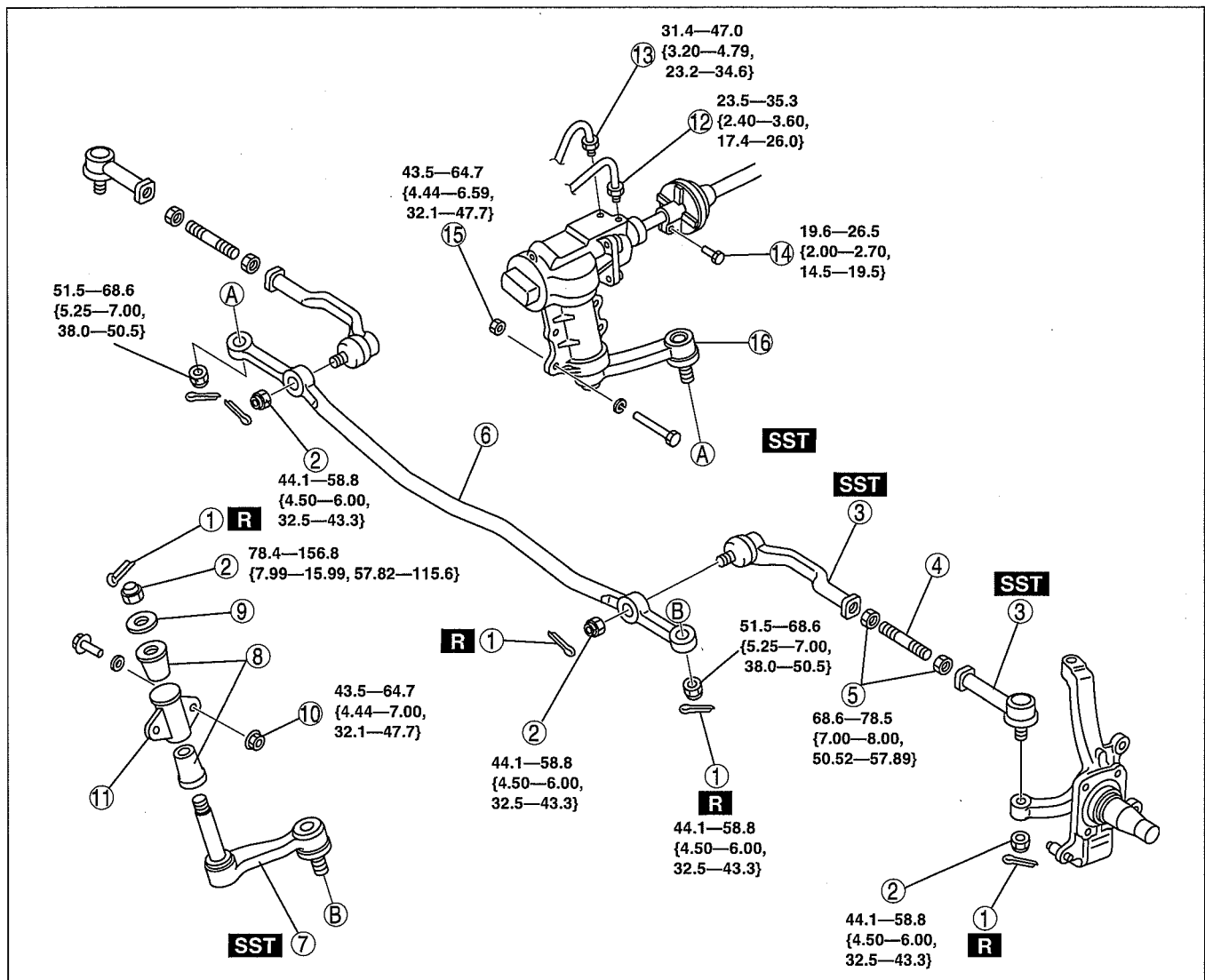
STEERING GEAR AND LINKAGE REMOVAL/INSTALLATION

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Caution

- Performing the following procedures without first removing the ABS wheel-speed sensor may possibly cause an open circuit in the harness if it is pulled by mistake. Before performing the following procedures, remove the ABS wheel-speed sensor (axle side) and fix it to an appropriate place where the sensor will not be pulled by mistake while servicing the vehicle.

- Remove the front ABS wheel-speed sensor. (with 4W-ABS) (See 04-13B-7 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))].) (See 04-13B-8 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)].)
- Remove in the order indicated in the table.
- Install in the reverse order of removal.
- After installation, adjust the total toe-in. (See 02-11-3 FRONT WHEEL ALIGNMENT [4x2].) (See 02-11-5 FRONT WHEEL ALIGNMENT [4x4].)



DBR614ZWB502

1	Cotter pin
2	Locknut
3	Ball joint (See 06-14-11 Ball Joint, Idler Arm, Steering Gear and Pitman Arm Removal Note.)
4	Tie rod
5	Locknut
6	Center link
7	Idler arm (See 06-14-11 Ball Joint, Idler Arm, Steering Gear and Pitman Arm Removal Note.)

8	Bushing
9	Washer
10	Bolt, Nut, washer
11	Idler arm bracket
12	Pressure pipe
13	Return pipe
14	Bolt
15	Bolt, Nut, Washer
16	Steering gear and pitman arm (See 06-14-11 Ball Joint, Idler Arm, Steering Gear and Pitman Arm Removal Note.)

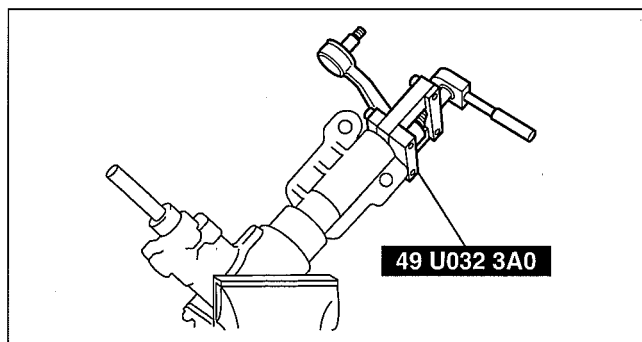
POWER STEERING

9	Oil seal
10	Y packing
11	Back up ring
12	Valve and piston component (See 06-14-12 Valve and Piston Component Assembly Note.)

13	O-ring
14	Seal ring
15	Gear housing

Pitman Arm Disassembly Note

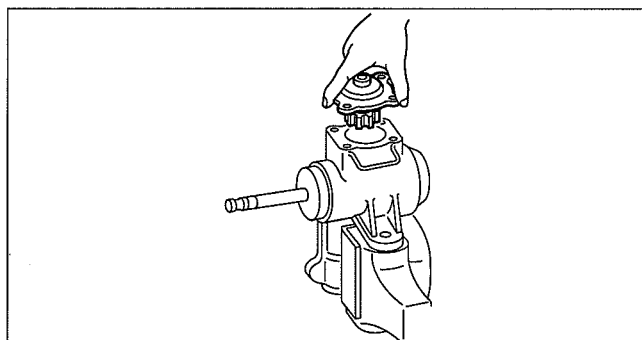
1. Separate the pitman arm from the steering gear using the SST.



DBR614ZWB008

Locknut, Sector Shaft Disassembly Note

1. Loosen the locknut.
2. Remove the side cover attaching bolts.
3. Set the sector shaft in the middle position.
4. Tap the sector shaft lower end using a plastic hammer to loosen the sector shaft.
5. Lift and remove the sector shaft with the locknut and the side cover from the gear housing.



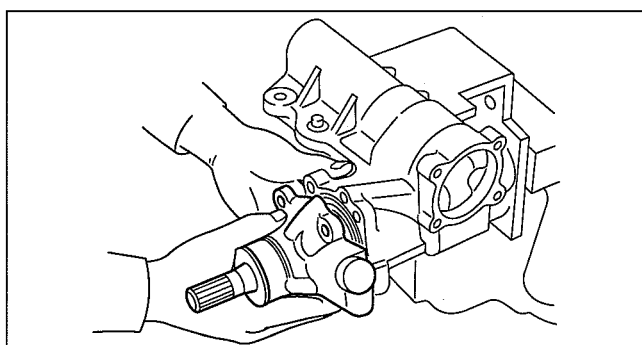
DBR614ZWB009

Valve and Piston Component Assembly Note

Caution

- Do not scratch the seal ring and O-ring on the piston against the entrance to the housing
- Insert the piston by slightly turning to the left and right so as not to create flexion at the O-ring and seal ring.

1. Insert the valve and piston component to the gear housing.



DBR614ZWB019

POWER STEERING

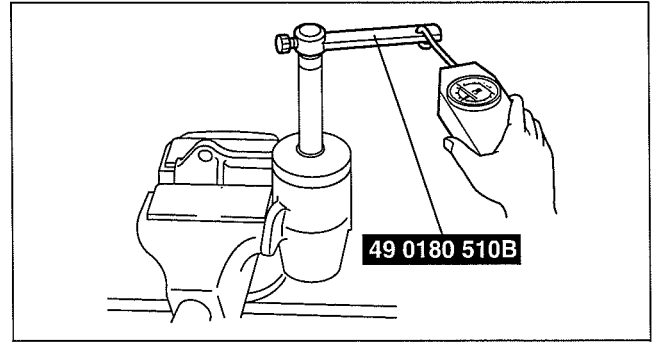
2. Inspect the worm shaft preload using the SST.

Steering gear worm shaft preload

0.64—1.23 N·m {6.6—12.5 kgf·cm, 5.7—10.8 in·lbf}

[Pull scale reading: 6.4—12.3 N {0.66—1.25 kgf, 1.44—2.76 lbf}]

- If not within specification, replace the valve and piston component.



DBR614ZWB898

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STEERING GEAR AND LINKAGE INSPECTION

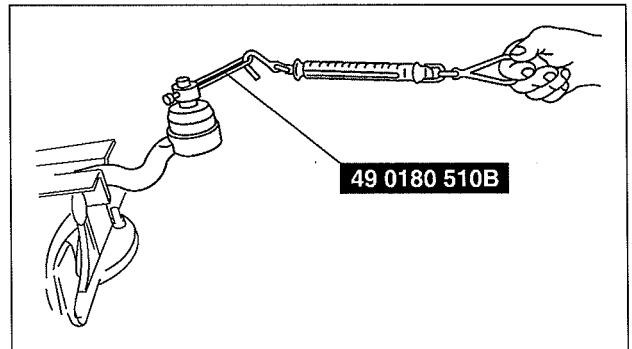
Tie-rod End Inspection

1. Inspect the tie-rod end for damage and excessive play.
 - If there are any cracks, replace the tie-rod end.
2. Inspect the boot for cracks.
 - If there is any malfunction, replace the tie-rod end boot.
3. Rotate the ball joint **5 times**.
 - If there is any malfunction, replace the tie-rod end boot.
4. Measure the rotational torque of the tie-rod end using the **SST** and a pull scale.
 - If not within the specification, replace the tie-rod end.

Tie-rod end rotational torque

0.49—1.18 N·m {5—12 kgf·cm, 4.4—10.4 in·lbf}

[Pull scale reading: 4.9—11.8 N·m {0.5—1.2 kgf, 1.11—2.65 lbf}]



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Steering Gear Backlash Adjustment

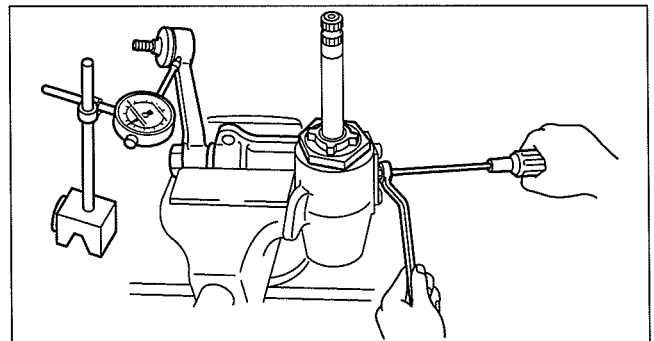
Caution

- Adjust the backlash while keeping the steering gear in the center position. Otherwise, the backlash becomes excessively small and gears may be damaged.

1. Turn the adjusting screw to adjust steering gear backlash.

Steering gear backlash

0 mm {0 in}



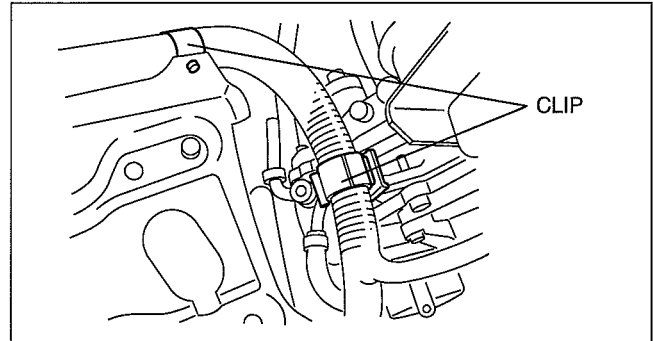
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POWER STEERING

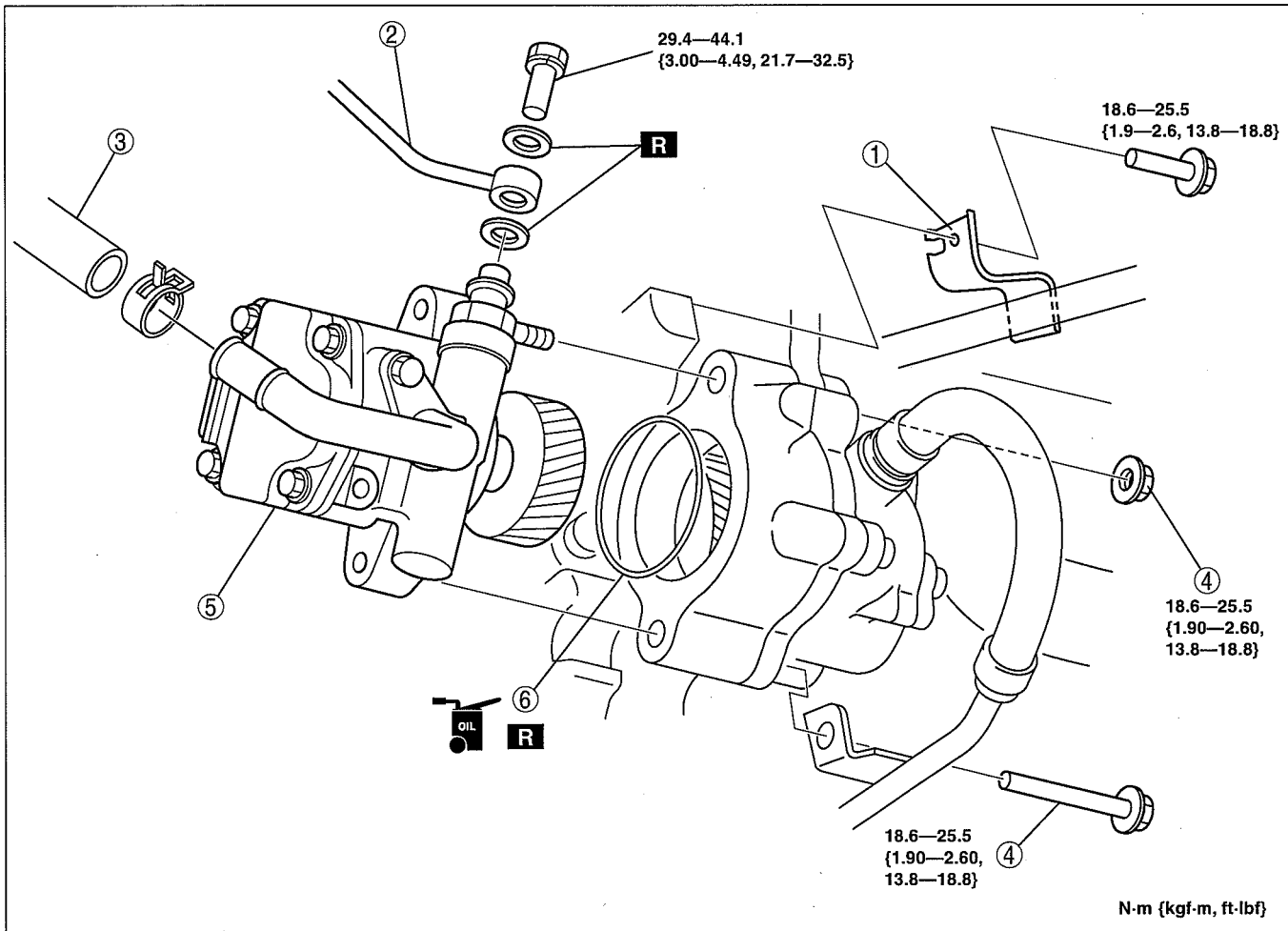
POWER STEERING OIL PUMP REMOVAL/INSTALLATION

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1. Remove the battery and battery tray. (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)
2. Remove the clip shown in the figure.
3. Remove in the order indicated in the table.
4. Install in the reverse order of removal.



DBR614ZWB020



DBR614ZWB121

1	Bracket (except WL-3)
2	Pressure pipe
3	Return hose

4	Bolt and nut
5	Power steering oil pump
6	O-ring

POWER STEERING

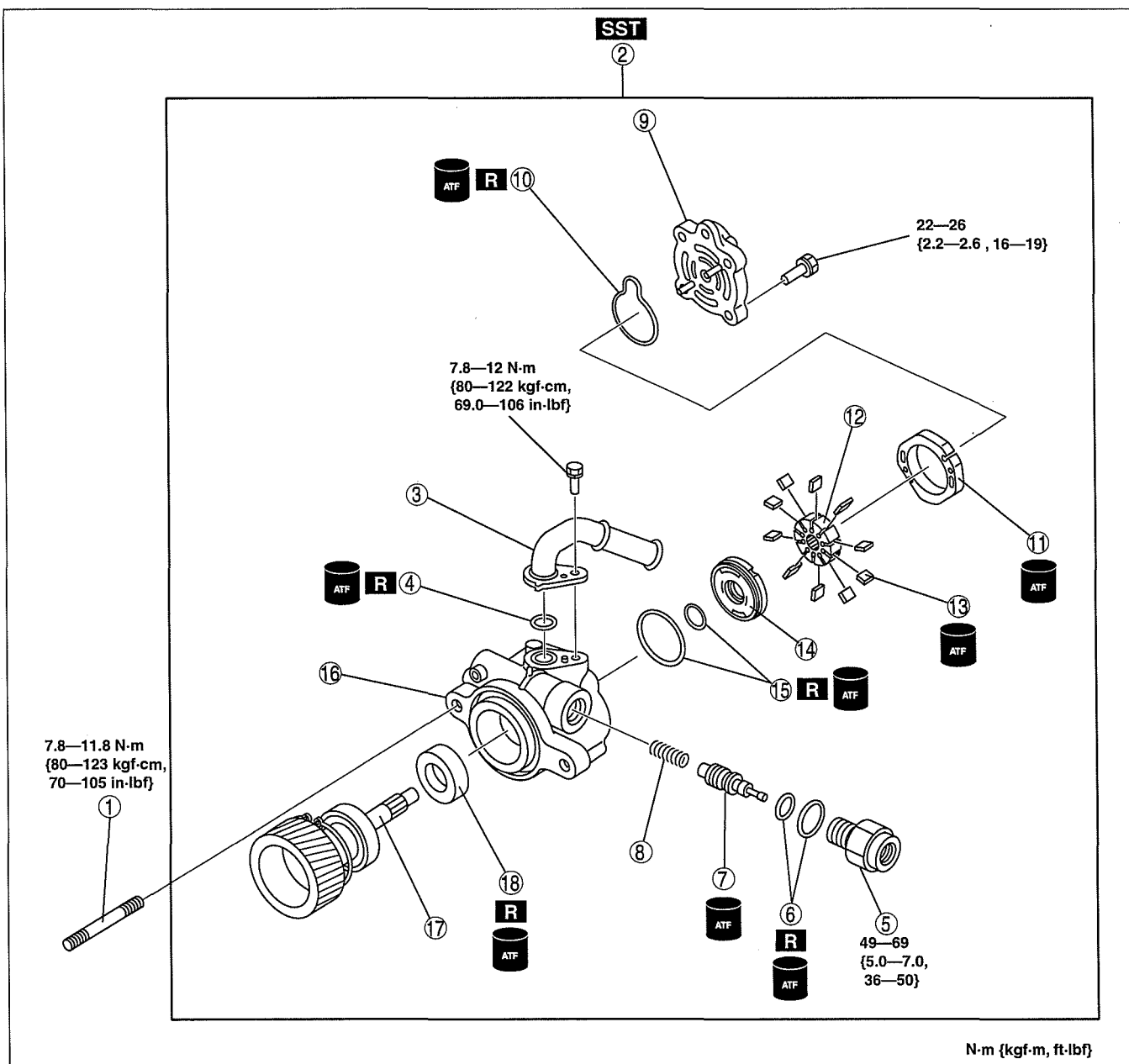
POWER STEERING OIL PUMP DISASSEMBLY/ASSEMBLY

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Note

- The following procedure is for replacement of the O-ring and oil seal only. Replace the pump component if other repairs are necessary.

- Disassemble in the order indicated in the table.
- Assemble in the reverse order of disassembly.
- After installing the P/S pump, flush the power steering system according to the following procedures.
 - Add ATF until it reaches the MAX level of the P/S fluid tank.
 - Start the engine and turn the steering wheel fully to the left and right several times to circulate the ATF.
 - If the fluid level is dropped, add ATF
 - Drain the ATF.
 - Add the new ATF to the MAX level of the P/S fluid tank, and repeat Step (1) and (2) a few times.



N-m {kgf-m, ft-lbf}

DBR614ZW901

1	Stud (See 06-14-16 Stud Disassembly Note.) (See 06-14-19 Stud Assembly Note.)
2	Power steering oil pump component (See 06-14-16 Power Steering Oil Pump Component Disassembly Note.)

3	Suction pipe
4	O-ring
5	Connector
6	O-ring
7	Control valve

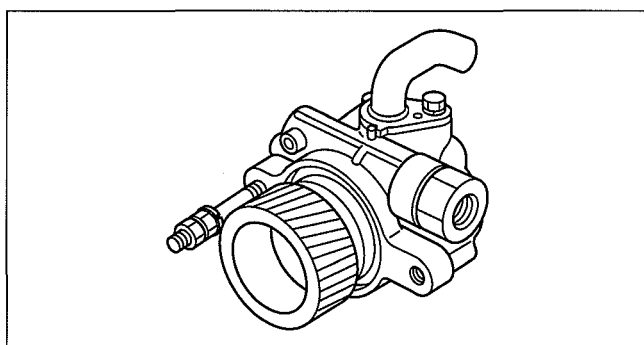
POWER STEERING

8	Spring
9	Rear pump body (See 06-14-18 Rear Pump Body Assembly Note.)
10	O-ring
11	Cam ring (See 06-14-18 Rotor, Cam Ring, Vane, Assembly Note.)
12	Rotor (See 06-14-18 Rotor, Cam Ring, Vane, Assembly Note.)
13	Vane (See 06-14-18 Rotor, Cam Ring, Vane, Assembly Note.)

14	Side plate (See 06-14-18 Side Plate Assembly Note.)
15	O-rings
16	Front pump body
17	Gear shaft component (See 06-14-16 Gear Shaft Component Disassembly Note.) (See 06-14-17 Gear Shaft Component Assembly Note.)
18	Oil seal (See 06-14-17 Oil Seal Disassembly Note.) (See 06-14-17 Oil Seal Assembly Note.)

Stud Disassembly Note

1. Tighten 2 nuts against each other on the stud, and then remove it.



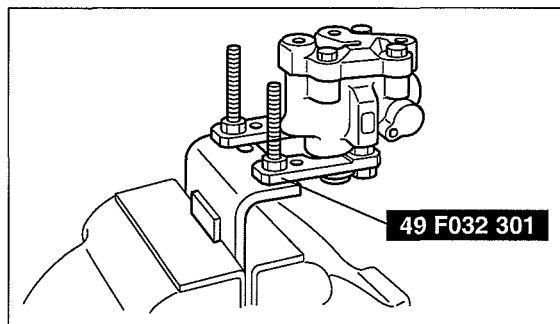
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Power Steering Oil Pump Component Disassembly Note

1. Secure the power steering oil pump using the SST.

Caution

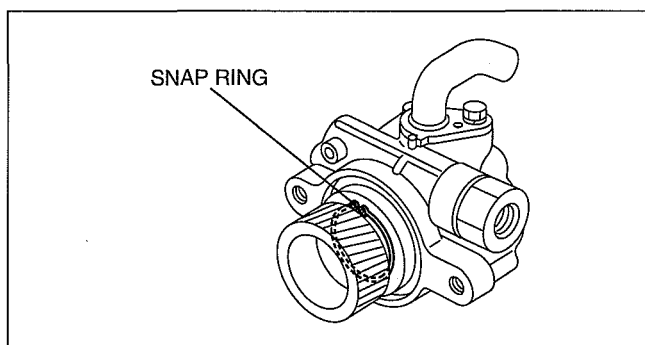
- Use the SST to prevent damage to the pump when securing it in a vise.



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Gear Shaft Component Disassembly Note

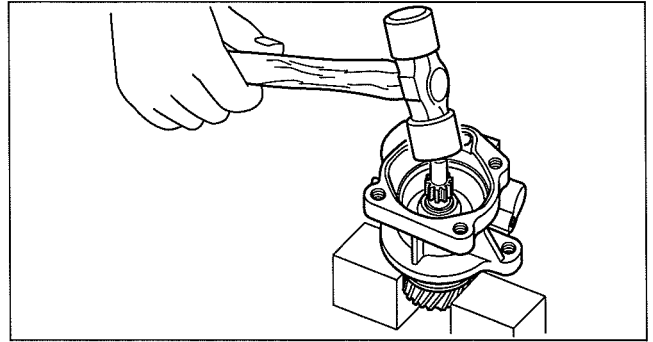
1. Remove the snap ring to front body shown in the figure.



DBR614ZWB888

POWER STEERING

2. Tap the gear shaft from the shaft side using a plastic hammer to remove it.



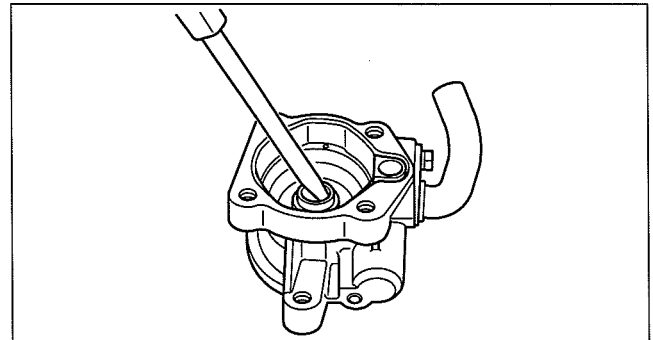
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Oil Seal Disassembly Note

1. Tap the oil seal out using a flathead screwdriver.

Caution

- To prevent damaging the power steering oil pump body when tapping it out, wrap a clean rag on the end of the flathead screwdriver. If the power steering oil pump body is damaged, replace it together with the power steering oil pump component.



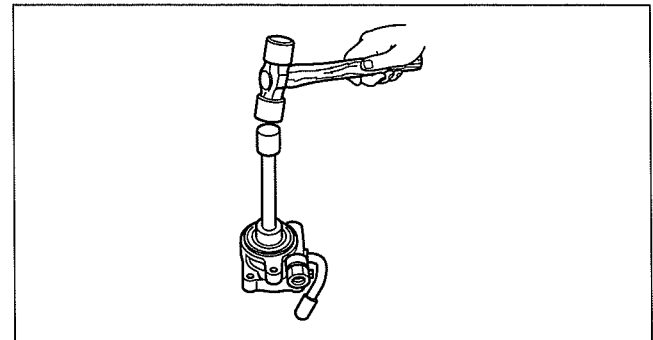
DBR614ZWB204

Oil Seal Assembly Note

1. Press fit the oil seal using a proper pipe.

Caution

- Install it so that the side with narrow groove points outwards.

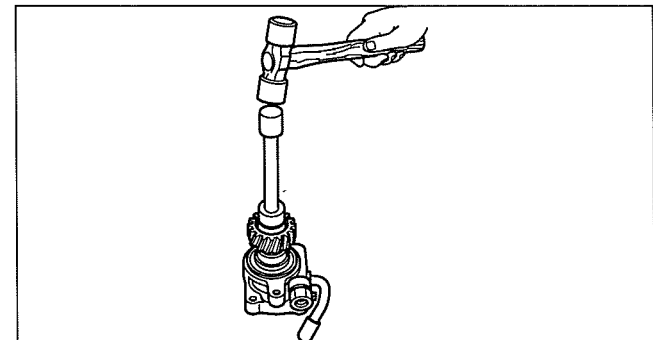


DBR614ZWB205

06

Gear Shaft Component Assembly Note

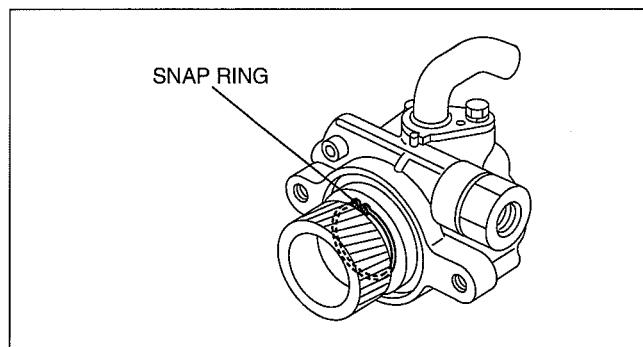
1. Set a proper pipe into the gear depression and press fit the gear shaft.



DBR614ZWB207

POWER STEERING

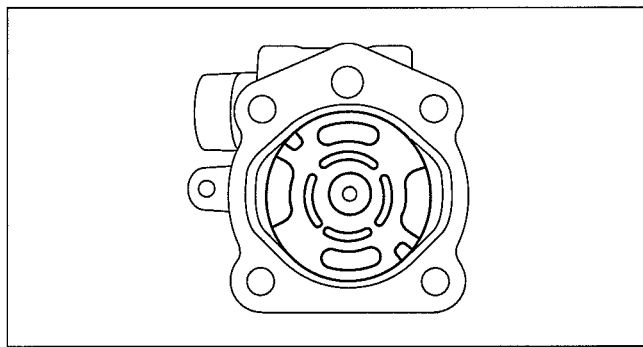
2. Install the snap ring to front body shown in the figure.



DBR614ZWB888

Side Plate Assembly Note

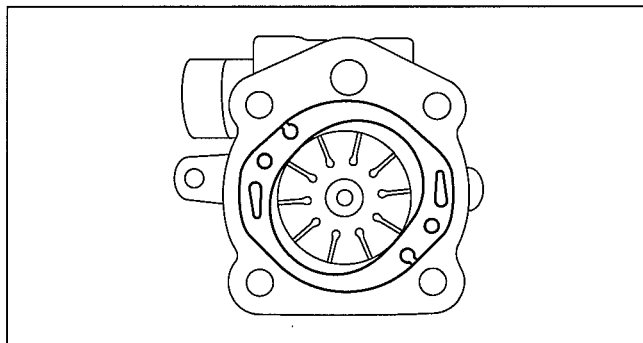
1. Install the side plate in the front pump body shown in the figure



DBR614ZWB500

Rotor, Cam Ring, Vane, Assembly Note

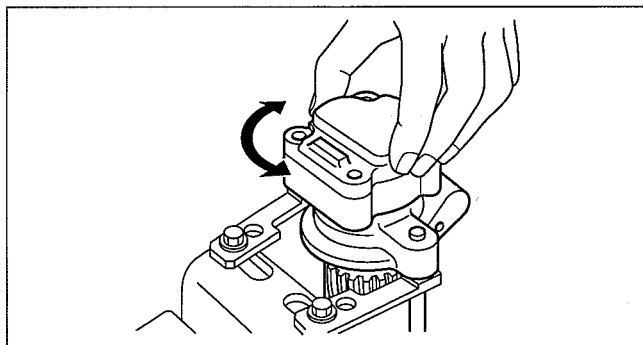
1. Install the rotor in the front pump body shown in the figure.
2. Install the cam ring in the front pump body shown in the figure.
3. Install the vane in the rotor.



DBR614ZWB206

Rear Pump Body Assembly Note

1. Install a new O-ring and the side plate to the front body.
2. Position the side plate using the rear pump body.
3. Install the rear body to the rear pump body.

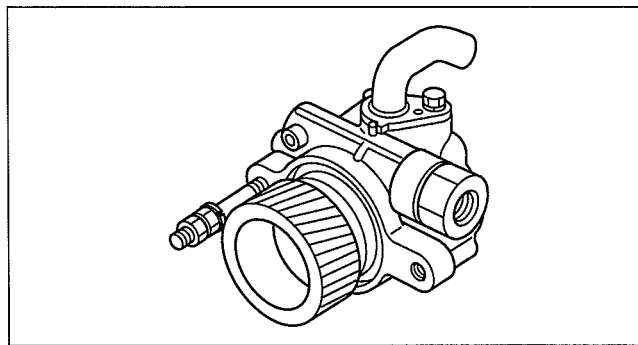


DBR614ZWB208

POWER STEERING

Stud Assembly Note

1. Tighten 2 nuts against each other on the stud, and then install it so that the stud projection is within the following specification.



DBR614ZWB201

TECHNICAL DATA

06-50 TECHNICAL DATA

STEERING TECHNICAL DATA..... 06-50-1

STEERING TECHNICAL DATA

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Item	Specification
Steering wheel play	0—30 mm {0—1.18 in}
Steering wheel effort	7.8 N·m {80 kgf·cm, 58 in·lbf} max.
Steering shaft length	789.9—792.9 mm {31.10—31.22 in}
Steering gear worm shaft preload	0.64—1.23 N·m {6.6—12.5 kgf·cm, 5.7—10.8 in·lbf} [Pull scale reading: 6.4—12.3 N {0.66—1.25 kgf, 1.44—2.76 lbf}]
Tie-rod end rotational torque	0.49—1.18 N·m {5—12 kgf·cm, 4.4—10.4 in·lbf} [Pull scale reading: 4.9—11.8 N·m {0.5—1.2 kgf, 1.11—2.65 lbf}]
Steering gear backlash	0 mm
Power steering fluid type	ATF M-III or equivalent (e.g. Dexron®II)
Power steering fluid capacity (approx. quantity)	1.1 L {1.2 US qt, 1.0 Imp qt}
Oil pump fluid pressure	9.3—9.8 MPa {94.8—99.9 kgf/cm ² , 1348—1421 psi}
Gear housing fluid pressure	9.30—9.79 MPa {94.8—99.8 kgf/cm ² , 1348—1419psi}

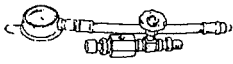
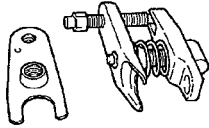
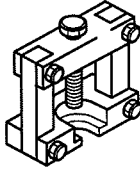
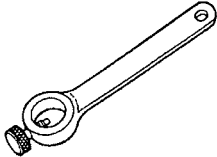
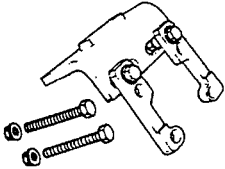
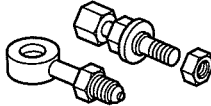
SERVICE TOOLS

06-60 SERVICE TOOLS

STEERING SST..... 06-60-1

STEERING SST

dcf06600000w01

49 1232 670B Power steering gauge set 	49 H002 671 Adapter 	49 G032 3A4A Power steering gauge adapter set 
49 T028 3A0 Ball joint puller set 	49 U032 3A0 Pitman arm 	49 0180 510B Preload measuring attachment 
49 F032 301 Power Steering Pump Hanger 	49 B032 304 Adapter 	49 G032 354 Adjusting Wrench 